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Appendix B — Extended Results

online part

Online supplement to the Master's thesis

Flow Matching Predictive Control with Constraints

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This file holds the three sections of Appendix B that are not printed in the thesis, with their tables, typeset as in the thesis: Section B.6, *UAV-corridor under Every Selection Rule* (Tables B.2 to B.5); Section B.8, *The Selection Rule against a Single Candidate Plan* (Table B.6); and Section B.9, *D3IL-avoiding at Twenty Episodes per Seed* (Tables B.7 and B.8). Section and table numbers are those of the thesis. A reference to a chapter, section, table or figure that is not in this file points into the thesis.

The code of the thesis is released at <https://github.com/ghubliming/fmpcc>; this file is in its folder data.

B.6. UAV-corridor under Every Selection Rule

Table 6.10 prints each model at each budget at its best selection rule under each projection method: the rule with the most successes, then the fewest violating steps, then the least time. Where a cell's flights all traverse the corridor, the three rules of a flow-based model lie within about two violating steps per flight (2.1 at most) and within about twenty control steps (21.4 at most) of one another, cumulative projection cost leaving the fewest violating steps in most cells. Under per-step projection over the hump at $K = 1$ no rule of any flow-based model traverses, and at $K = 2$ only temporal consistency does, on one flight of ten for analytic average-velocity matching, two for the consistency-interpolated model and none for instantaneous-velocity matching; the zero violating steps of the other rules there belong to flights that stall short of the apex until the step cap, not to constrained traversals. The baseline's rules differ by about five violating steps under the tilt, 0.8 under cumulative projection cost against 6.1 under temporal consistency, and its cumulative-cost rule over the hump traverses on nine flights of ten. The four tables below (Table B.2 to Table B.5) print every rule, per constraint and projection method. A flight succeeds when it crosses the finish line at the end of the corridor in controlled flight: the crossing is latched when it happens, and the controlled-flight checks of Subsection 5.4.3 are taken over the whole flight; the constraint is read as violating control steps per flight.

B. Extended Results

Table B.2.: UAV-corridor, *tilt*, per-step projection of Diffusion Predictive Control with Constraints (DPCC) under each selection rule (*r* random, *c* cumulative projection cost, *t* temporal consistency; four candidate plans), every budget of the grid; tightened constraints, activation threshold 0.5, ten flights per cell, one training seed. Columns as in Table 6.10, each mean $\pm$ the sample standard deviation over the flights. At the route's goal point the baseline's success is 0.200, 0.000 and 0.300. • the diffusion model of DPCC.

Model	K	Rule	Success	Violating steps	Steps	ms/step
Analytic-MeanFM	1	<i>r</i>	1.000 $\pm$ 0.000	10.4 $\pm$ 1.1	260.2 $\pm$ 4.9	28.5 $\pm$ 0.8
Analytic-MeanFM	1	<i>c</i>	1.000 $\pm$ 0.000	10.2 $\pm$ 1.1	253.2 $\pm$ 4.4	27.4 $\pm$ 0.5
Analytic-MeanFM	1	<i>t</i>	1.000 $\pm$ 0.000	10.3 $\pm$ 1.1	249.2 $\pm$ 5.0	28.7 $\pm$ 1.0
Analytic-MeanFM	2	<i>r</i>	1.000 $\pm$ 0.000	11.0 $\pm$ 1.2	262.1 $\pm$ 4.4	37.6 $\pm$ 0.9
Analytic-MeanFM	2	<i>c</i>	1.000 $\pm$ 0.000	10.3 $\pm$ 0.9	253.9 $\pm$ 4.2	36.4 $\pm$ 0.8
Analytic-MeanFM	2	<i>t</i>	1.000 $\pm$ 0.000	10.6 $\pm$ 0.8	255.6 $\pm$ 5.1	37.6 $\pm$ 1.0
Analytic-MeanFM	3	<i>r</i>	1.000 $\pm$ 0.000	7.5 $\pm$ 0.8	257.1 $\pm$ 2.7	94.3 $\pm$ 1.5
Analytic-MeanFM	3	<i>c</i>	1.000 $\pm$ 0.000	6.6 $\pm$ 0.8	256.4 $\pm$ 2.9	93.6 $\pm$ 1.1
Analytic-MeanFM	3	<i>t</i>	1.000 $\pm$ 0.000	7.2 $\pm$ 1.2	261.4 $\pm$ 1.8	93.4 $\pm$ 1.3
CI-MeanFM	1	<i>r</i>	1.000 $\pm$ 0.000	10.5 $\pm$ 1.2	259.2 $\pm$ 3.5	29.4 $\pm$ 0.8
CI-MeanFM	1	<i>c</i>	1.000 $\pm$ 0.000	10.3 $\pm$ 0.9	252.0 $\pm$ 3.7	27.9 $\pm$ 0.5
CI-MeanFM	1	<i>t</i>	1.000 $\pm$ 0.000	10.3 $\pm$ 1.3	247.2 $\pm$ 3.0	29.0 $\pm$ 0.8
CI-MeanFM	2	<i>r</i>	1.000 $\pm$ 0.000	10.5 $\pm$ 1.4	258.7 $\pm$ 4.4	37.6 $\pm$ 0.9
CI-MeanFM	2	<i>c</i>	1.000 $\pm$ 0.000	10.4 $\pm$ 0.7	251.9 $\pm$ 2.4	36.1 $\pm$ 0.3
CI-MeanFM	2	<i>t</i>	1.000 $\pm$ 0.000	10.6 $\pm$ 1.4	253.6 $\pm$ 4.6	37.3 $\pm$ 0.7
CI-MeanFM	3	<i>r</i>	1.000 $\pm$ 0.000	6.6 $\pm$ 1.0	255.3 $\pm$ 3.7	93.6 $\pm$ 1.8
CI-MeanFM	3	<i>c</i>	1.000 $\pm$ 0.000	5.0 $\pm$ 0.8	254.0 $\pm$ 1.8	93.4 $\pm$ 1.2
CI-MeanFM	3	<i>t</i>	1.000 $\pm$ 0.000	7.0 $\pm$ 1.2	258.0 $\pm$ 2.4	93.0 $\pm$ 1.2
FM	1	<i>r</i>	1.000 $\pm$ 0.000	10.2 $\pm$ 0.8	257.9 $\pm$ 9.6	28.1 $\pm$ 2.0
FM	1	<i>c</i>	1.000 $\pm$ 0.000	10.2 $\pm$ 1.5	259.8 $\pm$ 9.2	27.6 $\pm$ 1.5
FM	1	<i>t</i>	1.000 $\pm$ 0.000	10.9 $\pm$ 1.0	256.9 $\pm$ 21.3	27.5 $\pm$ 1.2
FM	2	<i>r</i>	1.000 $\pm$ 0.000	10.4 $\pm$ 1.1	257.2 $\pm$ 8.4	34.8 $\pm$ 1.1
FM	2	<i>c</i>	1.000 $\pm$ 0.000	9.8 $\pm$ 1.4	253.9 $\pm$ 9.4	34.7 $\pm$ 0.5
FM	2	<i>t</i>	1.000 $\pm$ 0.000	11.1 $\pm$ 0.9	246.6 $\pm$ 12.3	34.6 $\pm$ 0.4
FM	3	<i>r</i>	1.000 $\pm$ 0.000	6.7 $\pm$ 1.5	247.1 $\pm$ 3.2	78.8 $\pm$ 3.2
FM	3	<i>c</i>	1.000 $\pm$ 0.000	5.8 $\pm$ 1.5	247.2 $\pm$ 3.4	79.2 $\pm$ 3.2
FM	3	<i>t</i>	1.000 $\pm$ 0.000	6.7 $\pm$ 1.2	241.1 $\pm$ 2.3	78.9 $\pm$ 2.9
FM	5	<i>r</i>	1.000 $\pm$ 0.000	5.4 $\pm$ 1.4	248.1 $\pm$ 2.0	117.8 $\pm$ 2.5
FM	5	<i>c</i>	1.000 $\pm$ 0.000	4.4 $\pm$ 1.5	247.6 $\pm$ 2.0	116.9 $\pm$ 2.4
FM	5	<i>t</i>	1.000 $\pm$ 0.000	5.3 $\pm$ 1.3	242.7 $\pm$ 1.3	116.8 $\pm$ 2.8
FM	20	<i>r</i>	1.000 $\pm$ 0.000	3.0 $\pm$ 0.9	253.3 $\pm$ 3.7	360.0 $\pm$ 1.1
FM	20	<i>c</i>	1.000 $\pm$ 0.000	2.1 $\pm$ 0.7	252.6 $\pm$ 2.7	360.4 $\pm$ 1.5
FM	20	<i>t</i>	1.000 $\pm$ 0.000	3.2 $\pm$ 0.8	247.7 $\pm$ 4.4	358.4 $\pm$ 1.3
• Diffusion	20	<i>r</i>	1.000 $\pm$ 0.000	3.2 $\pm$ 3.3	267.7 $\pm$ 37.7	679.7 $\pm$ 30.9
• Diffusion	20	<i>c</i>	1.000 $\pm$ 0.000	0.8 $\pm$ 1.8	290.3 $\pm$ 15.5	666.3 $\pm$ 16.7
• Diffusion	20	<i>t</i>	1.000 $\pm$ 0.000	6.1 $\pm$ 4.6	257.2 $\pm$ 45.7	650.6 $\pm$ 24.2

B. Extended Results

Table B.3.: UAV-corridor, *tilt*, endpoint projection with one candidate plan (*single*) and under the three rules over four, at the budgets at which it has a guiding step; tightened constraints, activation threshold 0.5, ten flights per cell, one training seed. Columns as in Table 6.10, each mean $\pm$ the sample standard deviation over the flights.

Model	K	Rule	Success	Violating steps	Steps	ms/step
Analytic-MeanFM	3	single	1.000 $\pm$ 0.000	9.2 $\pm$ 1.4	256.6 $\pm$ 4.7	46.2 $\pm$ 0.6
Analytic-MeanFM	3	<i>r</i>	1.000 $\pm$ 0.000	8.5 $\pm$ 0.8	257.3 $\pm$ 5.3	73.9 $\pm$ 0.5
Analytic-MeanFM	3	<i>c</i>	1.000 $\pm$ 0.000	7.6 $\pm$ 1.3	255.2 $\pm$ 4.5	73.5 $\pm$ 0.8
Analytic-MeanFM	3	<i>t</i>	1.000 $\pm$ 0.000	9.0 $\pm$ 1.2	261.6 $\pm$ 5.1	72.5 $\pm$ 0.2
CI-MeanFM	3	single	1.000 $\pm$ 0.000	8.8 $\pm$ 0.8	254.4 $\pm$ 5.0	46.2 $\pm$ 0.4
CI-MeanFM	3	<i>r</i>	1.000 $\pm$ 0.000	9.0 $\pm$ 0.9	254.6 $\pm$ 4.4	73.2 $\pm$ 0.6
CI-MeanFM	3	<i>c</i>	1.000 $\pm$ 0.000	7.1 $\pm$ 1.0	253.2 $\pm$ 3.5	72.7 $\pm$ 0.5
CI-MeanFM	3	<i>t</i>	1.000 $\pm$ 0.000	8.7 $\pm$ 1.5	259.2 $\pm$ 3.9	72.8 $\pm$ 0.2
FM	3	single	1.000 $\pm$ 0.000	8.0 $\pm$ 1.6	254.8 $\pm$ 3.2	43.8 $\pm$ 0.2
FM	3	<i>r</i>	1.000 $\pm$ 0.000	8.2 $\pm$ 1.5	256.6 $\pm$ 1.6	71.4 $\pm$ 0.5
FM	3	<i>c</i>	1.000 $\pm$ 0.000	6.9 $\pm$ 1.8	257.8 $\pm$ 3.3	71.6 $\pm$ 0.6
FM	3	<i>t</i>	1.000 $\pm$ 0.000	8.2 $\pm$ 1.4	248.2 $\pm$ 5.2	71.4 $\pm$ 0.9
FM	5	single	1.000 $\pm$ 0.000	6.0 $\pm$ 1.7	255.6 $\pm$ 4.2	73.6 $\pm$ 0.2
FM	5	<i>r</i>	1.000 $\pm$ 0.000	6.0 $\pm$ 1.4	257.4 $\pm$ 3.9	113.0 $\pm$ 0.7
FM	5	<i>c</i>	1.000 $\pm$ 0.000	4.9 $\pm$ 1.9	256.6 $\pm$ 2.9	115.2 $\pm$ 1.4
FM	5	<i>t</i>	1.000 $\pm$ 0.000	6.3 $\pm$ 1.3	250.3 $\pm$ 5.3	115.2 $\pm$ 0.3
FM	20	single	1.000 $\pm$ 0.000	2.3 $\pm$ 0.5	269.9 $\pm$ 10.5	288.9 $\pm$ 1.4
FM	20	<i>r</i>	1.000 $\pm$ 0.000	2.3 $\pm$ 0.9	269.3 $\pm$ 10.7	416.8 $\pm$ 1.4
FM	20	<i>c</i>	1.000 $\pm$ 0.000	1.4 $\pm$ 0.7	269.8 $\pm$ 10.5	416.9 $\pm$ 0.8
FM	20	<i>t</i>	1.000 $\pm$ 0.000	2.5 $\pm$ 0.7	260.5 $\pm$ 10.6	417.2 $\pm$ 0.6

B. Extended Results

Table B.4.: UAV-corridor, *hump*, per-step projection of DPCC under each selection rule (r random, c cumulative projection cost, t temporal consistency; four candidate plans), every budget of the grid; tightened constraints, activation threshold 0.5, ten flights per cell, one training seed. Columns as in Table 6.10, each mean $\pm$ the sample standard deviation over the flights. —: no flight reaches the end of the corridor. At the route’s goal point the baseline’s success is 0.300, 0.300 and 0.700. • the diffusion model of DPCC.

Model	K	Rule	Success	Violating steps	Steps	ms/step
Analytic-MeanFM	1	r	0.000 ± 0.000	0.0 ± 0.0	—	26.5 ± 0.2
Analytic-MeanFM	1	c	0.000 ± 0.000	0.0 ± 0.0	—	26.4 ± 0.0
Analytic-MeanFM	1	t	0.000 ± 0.000	0.0 ± 0.0	—	26.6 ± 0.2
Analytic-MeanFM	2	r	0.000 ± 0.000	0.0 ± 0.0	—	35.0 ± 0.4
Analytic-MeanFM	2	c	0.000 ± 0.000	0.0 ± 0.0	—	35.2 ± 0.3
Analytic-MeanFM	2	t	0.100 ± 0.316	17.9 ± 30.0	387.0	42.3 ± 10.9
Analytic-MeanFM	3	r	1.000 ± 0.000	5.9 ± 0.7	288.5 ± 3.7	88.9 ± 1.8
Analytic-MeanFM	3	c	1.000 ± 0.000	4.9 ± 0.9	286.3 ± 4.3	91.1 ± 1.8
Analytic-MeanFM	3	t	1.000 ± 0.000	6.2 ± 0.6	288.0 ± 4.6	89.8 ± 1.9
CI-MeanFM	1	r	0.000 ± 0.000	0.0 ± 0.0	—	26.7 ± 0.3
CI-MeanFM	1	c	0.000 ± 0.000	0.0 ± 0.0	—	26.7 ± 0.3
CI-MeanFM	1	t	0.000 ± 0.000	0.0 ± 0.0	—	26.6 ± 0.3
CI-MeanFM	2	r	0.000 ± 0.000	0.0 ± 0.0	—	35.4 ± 0.2
CI-MeanFM	2	c	0.000 ± 0.000	0.0 ± 0.0	—	35.6 ± 0.3
CI-MeanFM	2	t	0.200 ± 0.422	19.8 ± 32.1	388.0 ± 2.8	43.0 ± 12.2
CI-MeanFM	3	r	1.000 ± 0.000	5.0 ± 1.1	287.1 ± 4.1	89.6 ± 1.7
CI-MeanFM	3	c	1.000 ± 0.000	4.6 ± 0.8	282.7 ± 3.6	89.2 ± 1.9
CI-MeanFM	3	t	1.000 ± 0.000	6.5 ± 0.5	285.8 ± 2.9	89.2 ± 2.0
FM	1	r	0.000 ± 0.000	0.0 ± 0.0	—	26.3 ± 0.4
FM	1	c	0.000 ± 0.000	0.0 ± 0.0	—	26.2 ± 0.2
FM	1	t	0.000 ± 0.000	0.0 ± 0.0	—	26.3 ± 0.3
FM	2	r	0.000 ± 0.000	0.0 ± 0.0	—	34.5 ± 0.4
FM	2	c	0.000 ± 0.000	0.0 ± 0.0	—	34.6 ± 0.0
FM	2	t	0.000 ± 0.000	0.0 ± 0.0	—	34.7 ± 0.4
FM	3	r	1.000 ± 0.000	5.9 ± 0.3	278.7 ± 6.1	76.8 ± 2.7
FM	3	c	1.000 ± 0.000	5.6 ± 0.5	277.0 ± 7.3	77.1 ± 2.7
FM	3	t	1.000 ± 0.000	5.8 ± 0.4	269.5 ± 4.0	77.2 ± 2.5
FM	5	r	1.000 ± 0.000	4.2 ± 0.4	279.1 ± 4.6	113.2 ± 2.8
FM	5	c	1.000 ± 0.000	3.9 ± 0.7	277.8 ± 7.0	114.1 ± 2.3
FM	5	t	1.000 ± 0.000	4.3 ± 0.5	269.7 ± 3.5	112.1 ± 2.3
FM	20	r	1.000 ± 0.000	2.2 ± 0.8	286.4 ± 3.5	354.3 ± 2.6
FM	20	c	1.000 ± 0.000	1.8 ± 0.4	285.1 ± 5.6	355.2 ± 2.2
FM	20	t	1.000 ± 0.000	2.2 ± 0.4	277.2 ± 2.3	356.3 ± 2.7
• Diffusion	20	r	1.000 ± 0.000	0.0 ± 0.0	320.8 ± 53.6	654.2 ± 89.7
• Diffusion	20	c	0.900 ± 0.316	0.0 ± 0.0	345.8 ± 22.8	623.1 ± 87.8
• Diffusion	20	t	1.000 ± 0.000	0.4 ± 1.3	308.8 ± 47.4	625.9 ± 75.0

B. Extended Results

Table B.5.: UAV-corridor, *hump*, endpoint projection with one candidate plan (*single*) and under the three rules over four, at the budgets at which it has a guiding step; tightened constraints, activation threshold 0.5, ten flights per cell, one training seed. Columns as in Table 6.10, each mean $\pm$ the sample standard deviation over the flights.

Model	K	Rule	Success	Violating steps	Steps	ms/step
Analytic-MeanFM	3	single	1.000 ± 0.000	10.1 ± 2.1	301.3 ± 4.6	45.9 ± 0.2
Analytic-MeanFM	3	r	1.000 ± 0.000	8.6 ± 2.5	303.0 ± 3.9	72.4 ± 0.6
Analytic-MeanFM	3	c	1.000 ± 0.000	8.0 ± 1.5	287.7 ± 4.9	72.1 ± 0.3
Analytic-MeanFM	3	t	1.000 ± 0.000	10.0 ± 0.8	291.3 ± 6.5	72.2 ± 0.3
CI-MeanFM	3	single	1.000 ± 0.000	10.4 ± 1.8	298.0 ± 4.1	45.6 ± 0.2
CI-MeanFM	3	r	1.000 ± 0.000	9.8 ± 1.9	298.0 ± 6.6	72.3 ± 0.7
CI-MeanFM	3	c	1.000 ± 0.000	8.0 ± 1.2	284.6 ± 2.8	71.9 ± 0.6
CI-MeanFM	3	t	1.000 ± 0.000	10.1 ± 1.2	288.8 ± 5.9	72.9 ± 0.7
FM	3	single	1.000 ± 0.000	10.1 ± 1.4	310.3 ± 7.7	44.2 ± 0.5
FM	3	r	1.000 ± 0.000	9.3 ± 1.3	310.2 ± 6.8	72.1 ± 0.8
FM	3	c	1.000 ± 0.000	8.3 ± 1.1	299.3 ± 4.7	71.9 ± 1.0
FM	3	t	1.000 ± 0.000	9.8 ± 0.8	288.8 ± 6.0	71.9 ± 1.2
FM	5	single	1.000 ± 0.000	4.9 ± 0.6	297.4 ± 4.1	72.9 ± 0.3
FM	5	r	1.000 ± 0.000	4.8 ± 0.6	296.7 ± 3.7	112.2 ± 0.5
FM	5	c	1.000 ± 0.000	3.9 ± 0.3	292.8 ± 3.7	112.4 ± 0.6
FM	5	t	1.000 ± 0.000	5.6 ± 0.5	283.6 ± 4.3	112.4 ± 0.7
FM	20	single	1.000 ± 0.000	0.2 ± 0.4	316.2 ± 6.3	291.0 ± 0.8
FM	20	r	1.000 ± 0.000	0.5 ± 0.7	317.3 ± 7.2	422.3 ± 1.2
FM	20	c	1.000 ± 0.000	0.4 ± 0.5	311.4 ± 4.9	424.6 ± 2.5
FM	20	t	1.000 ± 0.000	0.7 ± 0.7	301.8 ± 4.8	422.3 ± 0.9

B.8. The Selection Rule against a Single Candidate Plan

The random-rule rows of Chapter 6 behind the candidate-selection table of Section 8.2.

Table B.6.: The random rule (r), which executes one draw of the model and so stands for a single candidate plan, against the best of the three rules, for every configuration that Chapter 6 reports under all three; Table 8.1 gives the three groups. Per-step projection unless the model names endpoint projection; the UAV-corridor row covers both. *Measure*: S&C, success with constraint satisfaction (D3IL-avoiding, thirty episodes per cell unless stated); violation-free, contexts of ten without a violating control step (D3IL-aligning); crossings, flights of ten that cross the finish line, aborted flights in brackets (UAV-s-curve); violating steps per flight (UAV-corridor). Groups: the selection can be dropped, the random rule reaches the best rule; mixed, level on the measure but behind on a secondary one, or within a small margin; the selection is needed, the random rule falls short. ► the operating point of the task, • the baseline of record. The source of every row is in the last column.

Task	Model	K	Measure	r	Best rule	Source
<i>The selection can be dropped: the random rule reaches the best rule</i>						
D3IL-avoiding	FM	1	S&C	1.000	1.000 (c, t)	Table 6.2
D3IL-avoiding	FM	2	S&C	1.000	1.000 (c, t)	Table 6.2
D3IL-avoiding	CI-MeanFM	1	S&C	1.000	1.000 (t)	Table 6.2
D3IL-avoiding	Analytic-MeanFM	2	S&C	0.967	0.967 (t)	Table 6.2
D3IL-avoiding	Diffusion	2	S&C	1.000	1.000 (c)	Table 6.2
D3IL-aligning	Analytic-MeanFM, endpoint ►	20	violation-free	10/10	10/10 (c); t 9/10	Table 6.5
UAV-s-curve	FM	1	crossings	5/10 (3)	5/10 under every rule; aborts 3 (c), 5 (t)	Subsection 6.3.3
<i>Mixed: level on the measure, behind on a secondary one, or within a small margin</i>						
D3IL-aligning	Analytic-MeanFM	20	violation-free	9/10	9/10 (c); t 8/10 with 1.5 violating steps against 2.7	Table 6.5
UAV-corridor	flow-based	1–20	violating steps	—	the three rules within 2.1 per flight of one another where all flights traverse; c fewest in most cells	Section B.6
<i>The selection is needed: the random rule falls short</i>						
D3IL-avoiding	Analytic-MeanFM ►	1	S&C	0.900	0.967 (t)	Table 6.2
D3IL-avoiding	CI-MeanFM	2	S&C	0.900	1.000 (t)	Table 6.2
D3IL-avoiding	FM	20	S&C	0.867	1.000 (c, t)	Table 6.2
D3IL-avoiding	Diffusion •	20	S&C	0.967	1.000 (c)	Table 6.2
D3IL-avoiding	Analytic-MeanFM	10	S&C	0.833	0.933 (t)	Table 6.3

B.9. D3IL-avoiding at Twenty Episodes per Seed

Analytic average-velocity matching and instantaneous-velocity matching were evaluated a second time at the budgets they are operated at, $K = 1$ and $K = 2$, with twenty episodes per training seed and geometry in place of two: one hundred episodes per geometry and three hundred per configuration. Everything else is the configuration of Table 6.2: the same five trained checkpoints per model, four candidate plans, tightened constraints, the per-step projection of DPCC and the three selection rules. The comparison is within each model, the same configuration at two and at twenty episodes. Table B.7 prints it under projection and Table B.8 with projection switched off. These are the twenty-episode evaluations that are complete on all five training seeds and all three geometries. Consistency-interpolated average-velocity matching and the diffusion baseline have no twenty-episode evaluation that is complete in this sense, and at $K = 20$, where one control step takes about half a second (Table 6.2), none was completed for any model.

Success with constraint satisfaction. The two protocols agree. The operating point, analytic average-velocity matching at $K = 1$ under temporal consistency, reaches the goal within the constraints in 298 of 300 episodes at twenty episodes per seed and in 29 of 30 at two, and at $K = 2$ the counts are the same. Instantaneous-velocity matching under cumulative projection cost reaches 299 of 300 at $K = 1$ and 300 of 300 at $K = 2$, against 30 of 30 at both. In all sixteen configurations of the two tables, the count at two episodes lies inside the range of counts that 95 % of two-episode evaluations drawn from the twenty-episode one return.

What twenty episodes resolve. At two episodes per seed the count moves in steps of one episode in thirty, and the three selection rules of each model read within two episodes of each other. At twenty they separate. Analytic average-velocity matching under temporal consistency keeps 298 of 300 at both budgets, against 279 under random selection and 283 under cumulative projection cost at $K = 1$, and 280 and 278 at $K = 2$: the rule it is operated with is also the one with the highest success with constraint satisfaction. The three rules of instantaneous-velocity matching, all at 30 of 30 with two episodes, read 299, 299 and 297 of 300 at $K = 1$ and 299, 300 and 295 at $K = 2$. Drawn from these twenty-episode evaluations, a two-episode evaluation returns 30 of 30 with probability 0.81 for the operating point and 0.58 for instantaneous-velocity matching at $K = 2$ under temporal consistency. The spread over the five seeds narrows: for the operating point, the standard deviation of the success rate with constraint satisfaction falls from 0.075 at two episodes to 0.015 at twenty.

Control steps and time. The step counts behind Subsubsection 6.1.2.2 are reproduced. Under cumulative projection cost analytic average-velocity matching takes 72.0 control steps at $K = 1$ and 98.0 at $K = 2$, against 72.4 and 97.2 at two episodes, and under temporal consistency 61.0 and 60.4, against 58.6 and 59.4. Across the sixteen configurations the two protocols differ by at most 2.4 control steps. The time per control step of analytic average-velocity matching agrees to within 4 % under projection and 7 % without it; that of instantaneous-velocity matching is 5 to 13 % lower in the two-episode evaluation. The best rule of each model is the same at both protocols except for instantaneous-velocity matching at $K = 2$, whose three rules are level at two episodes and ordered cumulative projection cost, random selection, temporal consistency at twenty.

Without projection. Both models reach the goal in 297 to 299 of 300 episodes, against 30 of 30 at two episodes, and success with constraint satisfaction stays low: 36 of 300 for analytic average-velocity matching at both budgets and 15 and 30 of 300 for instantaneous-velocity matching at $K = 1$ and $K = 2$, against 1 and 3 of 30.

B. Extended Results

Table B.7.: D3IL-avoiding under per-step projection at twenty and at two episodes per training seed and geometry (columns 20 and 2): five training seeds $\times$ three geometries, 300 and 30 episodes per configuration; tightened constraints, four candidate plans, temporal U-Net of 4.0 M parameters. The two-episode columns are the cells of Table 6.2; rules as there. S&C is the number of episodes that reach the goal within the constraints. *Range*: the counts, of 30, that 95 % of two-episode evaluations return when the two episodes of each seed and geometry are drawn without replacement from its twenty, computed exactly from the fifteen seed-geometry counts of the twenty-episode evaluation. Control steps and time per control step are mean $\pm$ standard deviation over the five seeds (Subsection 5.4.4). $\blacktriangleright$ the operating point.

Model	K	Rule	S&C, episodes			Control steps		ms per control step	
			20	2	Range	20	2	20	2
Analytic-MeanFM	1	<i>r</i>	279/300	27/30	25–30	64.7 ± 1.8	64.3 ± 2.6	18.4 ± 0.1	19.2 ± 1.6
		<i>c</i>	283/300	28/30	26–30	72.0 ± 4.1	72.4 ± 8.4	18.0 ± 0.4	18.3 ± 0.5
		$\blacktriangleright t$	298/300	29/30	29–30	61.0 ± 1.0	58.6 ± 2.2	18.1 ± 0.3	18.3 ± 0.2
Analytic-MeanFM	2	<i>r</i>	280/300	29/30	25–30	65.9 ± 1.4	65.1 ± 2.6	27.3 ± 0.2	27.6 ± 0.5
		<i>c</i>	278/300	28/30	25–30	98.0 ± 10.4	97.2 ± 4.8	26.8 ± 0.4	27.1 ± 0.2
		<i>t</i>	298/300	29/30	29–30	60.4 ± 0.9	59.4 ± 2.0	27.1 ± 0.3	27.7 ± 0.5
FM	1	<i>r</i>	299/300	30/30	29–30	68.3 ± 0.9	67.9 ± 1.2	19.4 ± 0.6	17.4 ± 0.1
		<i>c</i>	299/300	30/30	29–30	67.5 ± 1.2	67.0 ± 0.8	18.9 ± 0.2	17.3 ± 0.1
		<i>t</i>	297/300	30/30	29–30	67.1 ± 1.6	68.2 ± 1.7	20.1 ± 1.9	17.7 ± 0.6
FM	2	<i>r</i>	299/300	30/30	29–30	68.0 ± 1.6	65.7 ± 0.8	27.6 ± 0.5	26.2 ± 0.9
		<i>c</i>	300/300	30/30	30	68.4 ± 3.0	67.0 ± 3.8	27.3 ± 0.3	25.5 ± 0.2
		<i>t</i>	295/300	30/30	28–30	64.9 ± 1.9	63.8 ± 1.9	27.9 ± 0.5	25.9 ± 0.2

Table B.8.: The configurations of Table B.7 with projection switched off; the two-episode columns are the cells of Table 6.1. Success and S&C are counts of episodes, *Range* as in Table B.7, violating steps per episode as mean $\pm$ standard deviation over the five seeds.

Model	K	Success, episodes		S&C, episodes			Violating steps	
		20	2	20	2	Range	20	2
Analytic-MeanFM	1	297/300	30/30	36/300	1/30	1–7	14.5 ± 0.7	15.5 ± 2.1
Analytic-MeanFM	2	299/300	30/30	36/300	1/30	1–7	15.4 ± 0.9	15.5 ± 0.4
FM	1	297/300	30/30	15/300	3/30	0–4	17.3 ± 1.0	16.5 ± 3.4
FM	2	299/300	30/30	30/300	3/30	1–6	16.4 ± 1.3	15.6 ± 2.6